VII SEMESTER (PROFESSIONAL ELECTIVE-III)	L	T	P	C
	3	-	-	3
20AM7E01 - BLOCK CHAIN TECHNOLOGIES				

COURSE OBJECTIVES:

To understand block chain technology and Crypto currency works

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1: Demonstrate the block chain basics, Crypto currency.

CO2: To compare and contrast the use of different private vs. public block chain and use cases.

CO3: Design an innovative Bit coin Block chain and scripts, Block chain Science on varies coins.

CO4: Classify Permission Block chain and use cases – Hyper ledger, Corda.

CO5: Make Use of Block-chain in E-Governance, Land Registration, Medical Information Systems and others.

UNIT -I:

Introduction: Introduction, basic ideas behind block chain, how it is changing the landscape of digitalization, introduction to cryptographic concepts required, Block chain or distributed trust, Currency, Crypto currency, How a Crypto currency works, Financial services, Bitcoin prediction markets.

UNIT II:

Hashing, public key cryptosystems, private vs public block chain and use cases, Hash Puzzles, Extensibility of Block chain concepts, Digital Identity verification, Block chain Neutrality, Digital art, Block chain Environment

UNIT III:

Introduction to Bitcoin: Bitcoin Block chain and scripts, Use cases of Bitcoin Blockchain scripting language in micropayment, escrow etc Downside of Bit coin mining, Block chain Science: Grid coin, Folding coin, Block chain Genomics, Bit coin MOOCs.

UNIT IV:

Ethereum continued, IOTA, The real need for mining, consensus, Byzantine Generals Problem, and Consensus as a distributed coordination problem, Coming to private or permissioned block chains, Introduction to Hyper ledger, Currency, Token, Campus coin, Coin drop as a strategy for Public adoption, Currency Multiplicity, Demurrage currency

UNIT V:

Technical challenges, Business model challenges, Scandals and Public perception, Government Regulations, Uses of Block chain in E-Governance, Land Registration, Medical Information Systems.

Text Books:

1. Blockchain Blue print for Economy by Melanie Swan

Reference Books:

1. Blockchain Basics: A Non-Technical Introduction in 25 Steps 1st Edition, by Daniel Drescher.